

TARUN S

Computer Science Student | Python | AI/ML | Data Science

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PROFESSIONAL SUMMARY

Computer Science undergraduate at Amrita Vishwa Vidyapeetham with strong Python programming fundamentals and hands-on experience building AI/ML applications, RAG pipelines, and data analytics engines. Skilled in NumPy, Pandas, Scikit-learn, vector store databases, and RESTful APIs, with a strong problem-solving mindset and eagerness to build real-world intelligent solutions as a Python AI Engineer Intern at Baigen Techno Labs.

TECHNICAL SKILLS

- **PROGRAMMING:** Python (Strong), Java, JavaScript, C++, SQL
- **AI & MACHINE LEARNING:** Machine Learning, Supervised Learning, Retrieval-Augmented Generation (RAG), NLP, Vector Embeddings, Semantic Search, Model Evaluation
- **DATA SCIENCE & LIBRARIES:** NumPy, Pandas, Scikit-learn (TF-IDF, Text Classification), Data Preprocessing, Exploratory Data Analysis (EDA)
- **ML FRAMEWORKS & AI TOOLS:** Scikit-learn, ChromaDB, OpenAI APIs (Whisper, Embeddings), PyTorch & TensorFlow (Foundational)
- **WEB & BACKEND:** FastAPI, Django, REST APIs, PostgreSQL, SQLite, Node.js, Angular, React
- **TOOLS & CONCEPTS:** Git, GitHub, Docker, VS Code, Object-Oriented Programming (OOP), Agile Development

AI/ML & TECHNICAL PROJECTS

Spectrum AI — AI-Powered RAG & Document Intelligence Platform | Python, FastAPI, ChromaDB, OpenAI Whisper, RAG, Embeddings

- Engineered a Python AI learning platform utilizing OpenAI Whisper for speech-to-text audio transcription and multi-stage text chunking workflows.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline using ChromaDB vector database and semantic search embeddings (text-embedding-3-small) for context-aware Q&A.
- Developed automated data pipelines integrating speech transcription, vector storage, and LLM APIs to generate structured study notes, interactive quizzes, and dynamic knowledge graph visualizations.

HyperTrade AI — Predictive Market Analytics & Decision Engine | Python, NumPy, Pandas, Technical Indicator Algorithms, REST APIs

- Engineered a Python quantitative analytics engine computing technical indicators including RSI, 5/20-day Moving Averages (SMA), and price volatility metrics.
- Applied NumPy and Pandas for data preprocessing, time-series data cleaning, and real-time market data evaluation.
- Developed a multi-factor algorithmic scoring system, incorporating model training and evaluation, to deliver automated Buy/Hold/Sell recommendations with risk assessment.

Autonomous Email Management Agent — NLP & Text Analytics Engine | Python, JavaScript, n8n, Gmail API, Regex Parsing, Automation

- Designed an automated rule & text analysis engine processing email payloads to calculate priority scores and classify message categories.
- Applied natural language text analytics using keyword density scoring and regex matching to detect spam, parse attachments, and automate responses.

AI Collaboration & Meeting Platform | Python, Django, PostgreSQL, REST APIs

- Built backend RESTful services and relational database schemas in PostgreSQL for real-time collaboration and session management.
- Optimized database query performance and API endpoints to ensure seamless data transmission and high application reliability.

EDUCATION

B.Tech in Computer Science and Engineering (3rd Year | Expected Graduation: 2028)

Amrita Vishwa Vidyapeetham | Location: Chennai, India

Relevant Coursework: Machine Learning Foundations, Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems

EXPERIENCE & LEADERSHIP

Product & Support Lead — Early Stage Tech Startup (CLABROOM) | Chennai, India

- Front end developer intern at Quantum AI and electronics labs.
- **Vocal Bridge:** Worked with an early-stage **AI automation and voice-agent startup**, developing solutions for **automated voice interactions and business process automation**
- Managed technical feature iterations and customer issue resolution, bridging user feedback with backend engineering tasks.
- Collaborated with cross-functional development teams in agile sprint planning to accelerate deployment cycle.